MILITARY SPECIFICATION

READOUTS, INCANDESCENT TYPE,

GENERAL SPECIFICATION FOR

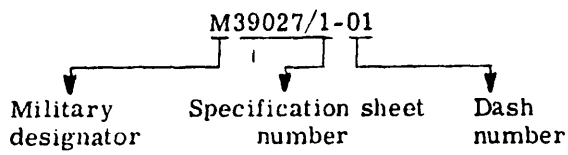
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for readouts, incandescent type, projection, backlighted (see 6.4) and segmented styles, single plane, consisting of single and multiple lamps, front relampable, with or without integral lenses.

1.2 Classification.

1.2.1 Military part number. The military part number shall consist of the letter "M", the basic number of the specification sheet, and an assigned dash number (see 3.1) as shown in the following:



1.2.2 Styles and classes. The style is identified by a two-letter symbol "RP" (Readout, projection), "RB" (Readout, backlighted), or "RS" (Readout, segmented), followed by a two-digit number (to be assigned when the specification sheet is issued).

Projection, style RP____

Class A - Viewing screen integral with unit.
Class B - Viewing screen separate with unit.
Class C - Viewing screen and bezel integral with unit.

Backlighted, style RB____

Class A - Viewing screen integral with unit.

Segmented, style RS_____

Class A - Slanted character.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

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SPECIFICATIONS

FEDERAL

L-P-410	-	Plastic, Polyamide, (Nylon), Rigid: Rods, Tubes, Flats, Molded and Cast Parts.
QQ-S-571	-	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
ZZ-R-765	-	Rubber, Silicone: Low- and High-Temperature and Tear Resistant.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Set-Up.
PPP-T-60	-	Tape; Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

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MIL-M-14	-	Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116	-	Preservation, Methods of.
MIL-L-3661	-	Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.
MIL-S-5059	-	Steel Corrosion, Resistant (18-8), Plate, Sheet and Strip.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-M-18012	-	Marking for Aircrew Station Displays, Design and Configuration of.
MIL-C-25050(ASG)	-	Colors, Aeronautical Lights and Lighting Equipment, General Specification For.
MIL-L-27160(USAF)	-	Lighting, Instrument, Integral, White, General Specification for.
MIL-G-45204	-	Gold Plating, Electrodeposited.
MIL-C-45662	-	Calibration System Requirements.
MIL-P-46120	-	Plastic Molding and Extrusion Material, Polysulfone.

(See Supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-3	-	Colors, Aeronautical Lighting.
FED-STD-406	-	Plastics, Method of Testing.
FED-STD-595	-	Colors.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-411	-	Aircrew Station Signals.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-749	-	Preparation and Submission of Data For Approval of Nonstandard Parts.
MIL-STD-783	-	Nomenclature and Abbreviations in Aircrew Stations.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Readout categories. Readouts furnished under this specification shall be category I, II, or III, as defined in 3.2.1, 3.2.2, or 3.2.3, respectively.

3.2.1 Category I readouts. Readouts completely defined by a military specification sheet (see 3.1 and 6.1.1).

3.2.2 Category II readouts. Readouts the same as category I readouts except for minor differences such as termination configuration and operating characteristics which do not change the basic design or construction of the qualified readout. Category II readouts shall be procured from a source listed on the applicable qualified products list for the particular similar product in category I. The specification sheet number for the similar product shall be specified (see 6.1.2). Category II readouts are non-standard 1/ (see 4.6.1 and 6.1.2).

3.2.3 Category III readouts. Readouts not covered by specification sheets. These readouts are non-standard 1/ (see 4.6.2 and 6.1.3).

3.3 Qualification. Readouts furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4.5 and 6.2).

3.4 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the readouts to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.4.1 Plastic material. Unless otherwise specified (see 3.1), thermoplastics shall not be used for fabricating readout cases and/or mounting hardware. When thermoplastic materials are used, they shall be self-extinguishing when tested in accordance with method 2021 or method 2022 of FED-STD-406 as applicable to the thinnest section of material used.

3.4.1.1 Parts within housings. If used for any parts within the assembly housings, the thermoplastic material shall be in accordance with L-P-410 or shall be of high temperature material which is self-extinguishing when tested in accordance with method 2021 of FED-STD-406.

1/ Test reports shall be prepared in accordance with MIL-STD-749.

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3.4.2 Lenses. Lenses shall be high temperature thermoplastic molding composition material.

3.4.3 Metals.

3.4.3.1 Metal parts. All metal parts, other than lamp contacts fabricated from solder, shall be of a corrosion resistant material or shall be suitably plated to resist corrosion.

3.4.3.2 Mounting hardware. Readout mounting hardware shall be made of a corrosion-resistant steel conforming to type 300 series of MIL-S-5059.

3.4.4 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal-spraying or metal plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals are defined in 6.3 through 6.3.4 and table VI, except that the use of corrosion resistant steel in contact with cadmium plating shall be allowed.

3.4.5 Rubber. All rubber parts shall be in accordance with ZZ-R-765.

3.4.6 Solder. Wherever suitable for the purpose, compositions Sn70, Sn60, and Sn5, conforming to QQ-S-571, shall be used. Solder of other compositions may be used, provided they possess superior characteristics and meet the requirements of this specification.

3.4.7 Finish. Unless otherwise specified (see 3.1), all external aluminum parts shall be anodized in accordance with MIL-A-8625. The exterior surfaces, designed to be exposed at the front of the panel after assembly, shall have a black lusterless finish, color 37038 in accordance with FED-STD-595.

3.5 Design and construction. Readouts shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 Housing. The housing shall be capable of independent or combination mounting as specified (see 6.1). Housing shall have an integral locking feature which facilitates front removal of lamps and positive locking to protect against vibration.

3.5.2 Plug-in units. The plug-in units shall be positive keyed to achieve proper alinement of legend in relation to fixed assembly when removed.

3.5.3 Screw threads. Screw threads shall be in accordance with requirement 12 of MIL-STD-454.

3.5.4 Contacts. Contacts shall be of a copper-based material and shall be suitably plated to meet all the applicable tests specified herein.

3.5.5 Contact springs. Contact springs shall be copper-beryllium alloy, suitably plated.

3.5.6 Terminals. Terminals shall be as specified (see 3.1 and 6.1).

3.5.6.1 Solder terminals. Solder terminals shall be treated to facilitate soldering. Coatings such as hot solder or hot-tin dip are acceptable. Gold plating shall not be used for solder-lug terminals.

3.5.6.2 Screw terminals. Screw terminals shall be provided with hardware as specified (see 3.1). Lockwashers shall be captive to the screw. For direct Government orders, all terminal hardware shall be assembled in proper order.

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3.5.6.3 Plug-in terminations (not applicable to printed circuit boards). Terminals shall be gold-plated in accordance with MIL-G-45204, type II, class 2, over nickel plate .001 inch to .003 inch thick.

3.5.7 Terminal marking. All lamp terminals shall be permanently marked (see 3.1). The marking shall remain legible at the completion of any and all tests specified herein.

3.5.8 Lamp. Unless otherwise specified (see 3.1), readouts shall be designed to accommodate T-1 wire terminal, T-1 flange base, T-1-3/4 midget flange base lamps or T-3-1/4 miniature bayonet base incandescent lamps. The number of lamps shall be as specified (see 3.1 and 6.1). The lamp circuits shall be isolated electrically from the applicable readout assembly housing.

3.5.9 Lamp replacement. Unless otherwise specified (see 3.1), the readouts shall be designed so that lamp replacement can be accomplished from the front of the panel without dismounting of the readout assembly housing from the panel and without the use of any tools.

3.6 Solderability (applicable to solder terminals). When readouts are tested as specified in 4.8.2, the dipped surface of the terminals shall be at least 95 percent covered with a new, smooth, solder coating. The remaining 5 percent of the terminal surface shall show only small pinholes or rough spots. These shall not be concentrated in one area. Bare base metal and areas where the solder dip failed to cover the original coating are indications of poor solderability, and shall be cause for failure. In case of dispute, the percent of coverage with pinholes or rough spots shall be determined by actual measurements of these areas, as compared to the total area.

3.7 Character display. When readouts are tested as specified in 4.8.3, the readout shall be capable of projecting on a single plane viewing screen, either individually or in combination a group of characters, words, messages, in colors as specified (see 6.1).

3.8 Character design and lighting color (see 6.5).

3.8.1 Character design and lettering shall be in accordance with MIL-M-18012. If legends are used they shall be in accordance with MIL-STD-783. When readouts are tested as specified in 4.8.4, the maximum character(s) height shall be as specified (see 3.1 and 6.1).

3.8.2 Color(s) when specified (see 6.1) shall be evenly distributed over the viewing screen when lighted. The view screen color(s) shall be in accordance with the following specifications:

(a) Level I colors.

	2200° Kelvin
Red	Type I (Aviation) per MIL-C-25050.
Yellow	Type I (Aviation) per MIL-C-25050.
Green	Type I (Aviation) per MIL-C-25050.
Blue	Category I (Aviation) per FED-STD-3.
White	White per MIL-L-27160.

(b) Level II colors.

	2200° Kelvin
Red	Type II (identification) per MIL-C-25050.
Yellow	Same as level I.
Green	Type II (identification) per MIL-C-25050.
Blue	Symbol B per MIL-L-3661.
White	Category I (Aviation) per FED-STD-3.

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(c) Level III colors.

Colors	Chromaticity limits 1/ 2100° Kelvin	
	X	Y
Red	0.665	0.335
	SL 2/	0.335
	SL 2/	SL 2/
	0.730	0.263
Yellow	SL 2/	0.450
	SL 2/	0.400
	0.590	0.400
	0.550	0.440
Green	0.200	0.650
	0.300	0.600
	0.350	0.500
	0.250	0.550
Blue	0.150	0.200
	0.200	0.200
	SL 2/	0.035
	SL 2/	0.125
White	0.450	0.420
	0.500	0.430
	0.500	0.395
	0.450	0.335

1/ The chromaticities of the diffusing viewing screens expressed as X and Y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using a source illuminate of 2100° Kelvin.
2/ SL - Spectrum locus (where intersected by other of coordinate pair).

3.9 Character distortion. When readouts are tested as specified in 4.8.5, the projected character(s) shall be clear and legible.

3.10 Display luminance. When readouts are tested as specified in 4.8.6, the brightness shall be in accordance with MIL-STD-411 and shall be as specified (see 3.1).

3.11 Field of view. When readouts, without an integral bezel, are tested as specified in 4.8.7, the projected characters shall be legible. Where the readout includes an integral bezel, the field of view shall be not less than described by a 40° included angle.

3.12 Thermal shock. When readouts are tested as specified in 4.8.8, there shall be no mechanical or electrical damage, loosening of rivets or other fastening devices or any discoloration or damage to the lenses.

3.13 Terminal strength. When readouts are tested as specified in 4.8.9, there shall be no movement or loosening of parts, visible permanent set, or other mechanical damage.

3.14 Mechanical endurance (applicable to plug-in-units only). When readouts are tested as specified in 4.8.10, there shall be no breakage of parts or degradation of performance. The removable plug-in-unit shall remain capable of removal by normal external pressure.

3.15 Dielectric withstanding voltage. When readouts are tested as specified in 4.8.11, there shall be no breakdown or flashover. The readouts shall remain operative at conclusion of the test.

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3.16 Shock. When readouts are tested as specified in 4.8.12, there shall be no chipping, cracking or crazing of materials, or loosening, bending, warping or distortion of parts. The plug-in units shall remain captive during the test and be capable of ejection by normal external pressure after the test.

3.17 Moisture resistance. When readouts are tested as specified in 4.8.13, there shall be no corrosion, decomposition, leaching out, separating of constituent materials, or other defects detrimental to the intended function of the readout. Immediately after the conclusion of the test and while the readouts are still in the humidity chamber, the insulation resistance shall be greater than 10 megohms. Following the 24-hour drying period, the insulation resistance shall be greater than 1,000 megohms.

3.18 Seal (when applicable) (see 3.1).

3.18.1 Dripproof. When readouts are tested as specified in 4.8.14.1, there shall be no leakage of water through the panel seal as determined by visual examination.

3.19 Vibration. When readouts are tested as specified in 4.8.15, there shall be no chipping, cracking, or crazing of materials, or loosening, bending, warping, or distortion of parts.

3.20 Salt spray (corrosion). When readouts are tested as specified in 4.8.16, there shall be no evidence of excessive corrosion. Excessive corrosion is defined as that which interferes with the electrical or mechanical performance, or in the case of plated metals, corrosion which has passed through the plating and attacked the base metal. There shall be no warping, cracking or other damage to the readouts.

3.21 Life (electrical). Upon completion of tests specified in 4.8.17, readouts shall meet requirements specified in 3.8 and 3.9.

3.22 Marking. Readouts shall be marked in accordance with method I of MIL-STD-1285 with the following information: 1/

(a) Manufacturer's trademark, or source code symbol.
(b) Military part number (see 1.2.1), or manufacturer's part number when not covered by a specification sheet.
(c) Date code.

3.23 Workmanship. Readouts shall be manufactured and processed in such a manner as to be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

1/ No trademarks, or other markings unrelated to the information displayed, shall appear on the screen.

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4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Qualification inspection (see 4.5).
(c) Inspection of readouts not covered by specification sheets (see 4.6).
(d) Quality conformance inspection (see 4.7).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the readouts, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable document
Plastic - - - - - - - - - - -	3.3.1	MIL-M-14, L-P-410, FED-STD-406
Metals - - - - - - - - - - -	3.3.4	MIL-S-5059
Rubber - - - - - - - - - - -	3.3.6	ZZ-R-765
Solder - - - - - - - - - - -	3.3.7	QQ-S-571
Finish - - - - - - - - - - -	3.3.8	MIL-A-8625, FED-STD-595

4.4 Inspection conditions. Unless otherwise specified herein, all inspection shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2) on sample units produced with equipment and procedures normally used in production.

4.5.1 Sample size. Thirty-four sample readouts of each style, assembled with the maximum number of readouts in an assembly for which qualification is sought, shall be submitted for qualification inspection. If more than one class is listed on the specification sheet, the samples shall be representative of all the classes for which qualification is sought.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into 3 groups of 3 sample units each and one group of 25 sample units and subjected to the inspection for their particular group.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.5.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 12 month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, group A, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) A summary of the results of test performed for qualification verification inspection, groups B and C, including the number and mode of failures. The summary shall include results of all qualification verification inspection tests performed and completed during the 12 month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

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Failure to submit the report within 30 days after the end of each 12 month period may result in loss of qualification for that product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 12 month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each style and class, to testing in accordance with the qualification inspection requirements.

TABLE II. Qualification inspection.

Examination or test 1/	Requirement paragraph	Method paragraph
Group I (all sample units)		
Visual and mechanical examination	3.1, 3.4 thru 3.5.9, 3.22 and 3.23	4.8.1
Solderability	3.6	4.8.2
Character display	3.7	4.8.3
Character design and lighting color	3.8	4.8.4
Character distortion	3.9	4.8.5
Display luminance	3.10	4.8.6
Field of view	3.11	4.8.7
Group II (3 sample units from group I)		
Thermal shock	3.12	4.8.8
Terminal strength	3.13	4.8.9
Mechanical endurance	3.14	4.8.10
Dielectric withstanding voltage	3.15	4.8.11
Group III (3 sample units from group I)		
Shock 2/	3.16	4.8.12
Moisture resistance	3.17	4.8.13
Character distortion	3.9	4.8.5
Seal (when specified)	3.18	4.8.14
Dielectric withstanding voltage	3.15	4.8.11
Group IV (3 sample units from group I)		
Vibration 2/	3.19	4.8.15
Mechanical endurance	3.14	4.8.10
Character distortion	3.9	4.8.5
Seal (when specified)	3.18	4.8.14
Dielectric withstanding voltage	3.15	4.8.11
Salt spray	3.20	4.8.16
Group V (25 sample units from group I)		
Life (electrical)	3.21	4.8.17

1/ If a single lamp failure occurs during any test specified, the test is discontinued and the lamp replaced and testing continued.

2/ Two additional sample units for shock and two addition sample units for vibration may be utilized to enable qualification for both shock methods and both vibration methods, respectively.

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4.6 Inspection requirements for category II and category III readouts (items not covered by specification sheets). Inspection requirements for items not covered by specification sheets shall be performed by the supplier, after award of contract, and prior to production. (See 6.2.2 and 6.2.3.)

4.6.1 Category II readouts. Additional tests to verify suitability of the variations from the category I readouts shall be performed as specified (see 6.2.2).

4.6.2 Category III readouts. Unless otherwise specified (see 6.2.3), the inspection requirements shall be as specified in 4.5 through 4.5.2 inclusive.

4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.7.1.1 Inspection lot. An inspection lot shall consist of all readouts of the same style, design, and construction, produced under essentially the same conditions and offered for inspection at one time. Similar readouts of different classes meeting these requirements but with different characters projected may be combined to form an inspection lot.

4.7.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, in the order shown.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination - - -	3.1, 3.4 thru 3.5.9, 3.22 and 3.23	4.8.1	1.0	4.0
Character height and lighting color - - -	3.8	4.8.4	1.0	---
Character distortion - - - - - - - - - - -	3.9	4.8.5	1.0	---

4.7.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

4.7.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.7.1.2.3 Disposition of sample units. Sample units which have passed all the group A inspection may be delivered on the contract or purchase order, if the lot is accepted.

4.7.2 Qualification verification inspection. Qualification verification inspection shall consist of groups B and C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.7.2.3), delivery of products which have passed group A shall not be delayed pending the results of these qualification verification inspections.

4.7.2.1 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table IV, in the order shown. Group B inspection shall be made on sample units which have passed the group A inspection.

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4.7.2.1.1 Sampling plan. Six sample units shall be subjected to group B inspection 12 months after the date of notification of qualification, and after each subsequent 12-month period. Group B inspection shall also be performed at any time after the date of notification of qualification, when production of a particular style of readout shall have been suspended for 6 months or more. In such a case, inspection shall be performed on sample units selected from the first lot of the new production presented for acceptance, and after each subsequent 12-month period. If production of the qualified item has been suspended for the previous 12 months, the qualifying activity shall be notified.

4.7.2.1.2 Failures. If one or more sample units fail to pass group B inspection, the sample shall be considered to have failed.

4.7.2.1.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

TABLE IV. Group B inspection.

Examination or test	Requirement paragraph	Method paragraph
Subgroup 1 (2 sample units)		
Solderability	3.6	4.8.2
Character display	3.7	4.8.3
Display luminance	3.10	4.8.6
Field of view	3.11	4.8.7
Thermal shock	3.12	4.8.8
Terminal strength	3.13	4.8.9
Mechanical endurance	3.14	4.8.10
Dielectric withstanding voltage	3.15	4.8.11
Subgroup 2 (2 sample units)		
Shock	3.16	4.8.12
Moisture resistance	3.17	4.8.13
Character distortion	3.9	4.8.5
Seal (when specified)	3.18	4.8.14
Dielectric withstanding voltage	3.15	4.8.11
Subgroup 3 (2 sample units)		
Vibration	3.19	4.8.15
Mechanical endurance	3.14	4.8.10
Character distortion	3.9	4.8.5
Seal (when specified)	3.18	4.8.14
Dielectric withstanding voltage	3.15	4.8.11
Salt spray	3.20	4.8.16

4.7.2.2 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table V, in the order shown. Group C inspection shall be made on sample units which have passed the group A inspection.

TABLE V. Group C inspection.

Examination or test	Requirement paragraph	Method paragraph
(25 sample units)		
Life (electrical)	3.21	4.8.17

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4.7.2.2.1 Sampling plan. Twenty-five sample units shall be selected from those covered by a single specification sheet. Group C inspection shall be performed 12 months after the date of notification of qualification, and after each subsequent 24-month period. Group C inspection shall also be performed at any time after the date of notification of qualification, when production of a particular style of readout shall have been suspended for 12 months or more. In such a case, inspection shall be performed on sample units selected from the first lot of the new production presented for acceptance, and after each subsequent 24-month period. If production of the qualified item has been suspended for the previous 24 months, the qualifying activity shall be notified.

4.7.2.2.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

4.7.2.2.3 Disposition of sample units. Sample units which have been subjected to group C inspection may be delivered on the contract or purchase order.

4.7.2.3 Noncompliance. If a sample fails to pass group B or C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B or C inspection, as applicable, shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B or C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7.3 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

4.8 Methods of examination and tests. All tests specified herein shall be conducted with lamps in place.

4.8.1 Visual and mechanical examination. Readouts shall be examined to verify that materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 thru 3.5.9, 3.22, and 3.23).

4.8.2 Solderability (see 3.6). Readouts shall be tested in accordance with method 208 of MIL-STD-202. The following details and exceptions shall apply:

(a) Number of termination - 4.
(b) Wire wrap and solder dipping machine - Not applicable.
(c) Examination of termination - Terminals shall have a 95 percent new, smooth solder coating. The remaining 5 percent shall have only small pinholes or rough spots, and shall not be concentrated in one area.

4.8.3 Character display (see 3.7). The characters shall be displayed on the integral viewing screen for all units except those which require an external viewing screen. For projection units that require an external screen, an acrylic plastic viewing screen having one side with diffusing surface shall be used. Where required, displayed characters shall be the numeral "2" of a size equivalent to the maximum specified size. Character display shall be as specified in 3.1.

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4.8.4 Character design and lighting color (see 3.8). The character(s) shall be displayed on a screen as described in 4.8.3. The displayed character(s) shall then be measured as specified in 3.8.

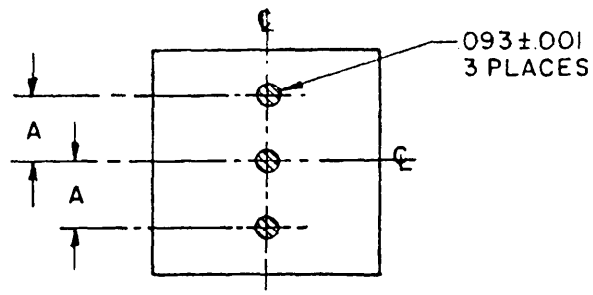
4.8.5 Character distortion (see 3.9). The character(s) shall be displayed on a screen as described in 4.8.3. The displayed characters shall then be measured while being viewed at a distance of 18 inches in a plane normal to the viewing screen.

4.8.6 Display luminance (see 3.10).

4.8.6.1 Display luminance for projection-type readouts. Three readings shall be taken with a Spectra spot brightness meter, or equivalent using an incandescent lamp, as specified (see 3.1) for each position using the mask in figure 1 and the average of the three readings shall be greater than that specified (see 3.1).

Example: 3 readings per position multiplied by number of positions as specified (see 3.1) equals cumulative photometer readings.

$$\frac{\text{Cumulative Photometer readings}}{\text{Total number of readings}} = \text{Average brightness}$$



NOTES:
1. Dimension 'A' is determined from the following relationships: $\frac{3 \times \text{useable message height} \pm .005}{8}$
2. Use front mask or projected dots.

FIGURE 1. Test projection mask.

4.8.6.2 Display luminance for segmented-type readouts. Three readings per segment shall be taken with a Spectra spot brightness meter, or equivalent using an incandescent lamp(s), as specified (see 3.1) for each segment. The average luminance of each bar segment shall be greater than that specified (see 3.1).

4.8.7 Field of view (see 3.11). The readout shall be mounted in a vertical position with the character(s) displayed on a screen as described in 4.8.3. The screen shall be viewed from a distance of 3 feet, unless otherwise specified (see 3.1), all angles up to 60° to a line perpendicular to the viewing surface.

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4.8.8 Thermal shock (see 3.12). Readouts shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

(a) Test condition letter - A.
(b) Measurements before and after cycling - Readouts shall be observed prior to and after cycling for conformance to 3.9 and 3.11.
(c) Examination after test - Readouts shall be examined for any mechanical or electrical damage and loosening of rivets or other fastening devices and discoloration or damage to lenses.

4.8.9 Terminal strength (see 3.13). Terminals of readouts shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition letter - As specified (see 3.1).
(b) Applied force - As specified (see 3.1).
(c) Following the test, readouts shall be examined for movement or loosening of parts, visible permanent set, or evidence of mechanical damage.

4.8.10 Mechanical endurance (applicable to plug-in units only) (see 3.14). Readouts shall be mounted by their normal mounting means and shall be subjected to 150 insertions and removals, either manually or automatically. After testing, the readouts shall be examined for compliance with 3.14.

4.8.11 Dielectric withstanding voltage (see 3.15). The dielectric withstanding voltage test shall be conducted in accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

(a) Dummy lamps which contain no filament shall be installed prior to test.
(b) Test voltage - 1,000 volts rms, projection units.
- 600 volts rms, backlighted units.
(c) Duration of application - 1 minute for qualification and group B inspection, 5 seconds for other tests.
(d) Points of application of test voltage are between mutually insulated portions, and between each insulated portion and exposed or grounded metal terminal(s).

4.8.12 Shock (see 3.16). Readouts shall be tested in accordance with 4.8.12.1 or when specified (see 3.1), in accordance with 4.8.12.2.

4.8.12.1 Method I. Readouts shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

(a) Special mounting means - Readouts shall be mounted on a rigid metal panel by their normal mounting means.
(b) Test condition letter - G.
(c) Electrical load conditions - Nonoperating.
(d) Examination after test - See 3.16.

4.8.12.2 Method II. Readouts assemblies shall be tested in accordance with method 207 of MIL-STD-202. The following details shall apply:

(a) Special mounting means - As specified in 4.8.12.1.
(b) Electrical load conditions - As specified in 4.8.12.1.
(c) Examination after test - As specified in 4.8.12.1.

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4.8.13 Moisture resistance (see 3.17). Readouts shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - By normal mounting means (see 3.1) on a corrosion-resistant metal panel extending beyond the readout device, positioned 15 degrees from vertical with readout legend pointing upward and uninsulated.
(b) Polarization - During steps 1 to 6 inclusive, a polarizing voltage of 100 volts dc shall be applied between all terminals tied together and the metal panel. Steps 7a and 7b are not applicable.
(c) Loading voltage - Not applicable.
(d) Final measurement - Insulation resistance. The insulation resistance test shall be in accordance with method 302 of MIL-STD-202, test condition letter B. Points of measurements shall be between each terminal and ground and between all adjacent terminals.

4.8.14 Seal (see 3.18).

4.8.14.1 Dripproof test. Before conducting the test, all gaskets which normally can be replaced in service without disassembly of the basic readout unit, shall be disassembled and reassembled with the readout unit. With the readout assembly mounted by its normal means and associated mounting hardware, the readout assembly shall be subjected to the dripproof test of MIL-STD-108 with a five gallon quantity of water flowing over the specimen for a period of five minutes duration. The water stream shall be directed within 12 inches above the test specimen. No louvers shall be installed between the item under test and the water flow.

4.8.15 Vibration (see 3.19). Readouts shall be tested in accordance with 4.8.15.1 or when specified (see 3.1) in accordance with 4.8.15.2.

4.8.15.1 Method 1. The readouts shall be tested in accordance with method 201 of MIL-STD-202. The following details shall apply:

(a) Tests and measurements prior to vibration - Not applicable.
(b) The readout shall be mounted in its related holding fixture without the use of clamps or support foreign to the structure. The holding fixture shall then be rigidly mounted to the shock vibration test platform.
(c) The readout shall support its maximum number of leads soldered to the terminals and extended in cable fashion to an anchor point remote from the item under test. The wire cable shall be of sufficient length in order not to restrict movement of the readout in its holding fixture during the test.
(d) Readouts shall be energized sequentially and monitored for intermittent light output during the test. Intermittent light output shall be detected by instrumenting the light output of a lamp in a second unit located off the vibrating platform with each of the two lamps, one on the platform and one off the platform, connected in a series circuit to provide a remote means of detecting performance of the device under test. The lamps used in each readout shall be of like characteristics in order to establish correlation of light output in respect to changes in the circuit constant between the two lamps. The readout used as a detector source shall not support film information. This readout shall be mounted in a fixed position within a light-tight enclosure in which the total light output on the viewing screen is measured by a suitable light meter. During the test, the measured light output at rated input voltage shall not change greater than ±25 percent from the light output measured at rated input voltage prior to the test. Any intermittence of the light level observed by the light meter that falls below the 25 percent initial light level shall also be cause for failure of the device under test.
(e) After completing the test, all readouts shall be examined for compliance with 3.7, 3.9, 3.11, and 3.16.

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4.8.15.2 Method II. Readouts shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting - As specified in 4.8.15.1.
(b) Electrical-load conditions - As specified in 4.8.15.1.
(c) Test condition letter - As specified (see 3.1).
(d) Measurements - As specified in 4.8.15.1.

4.8.16 Salt spray (see 3.20). Readouts shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply:

(a) Applicable salt solution - 5%.
(b) Test condition letter - B.
(c) Measurements after exposure - The readouts shall be examined for evidence of excessive corrosion, warping, cracking, or other damage.

4.8.17 Life (electrical) (see 3.21). Readouts shall be subjected to 300 hours of continuous operation under conditions of maximum operating power dissipation, that is, the maximum number of lamp(s) required to be energized for any one display character. The lamps shall be of the highest rated wattage, and shall be operated at the maximum lamp voltage. The following test procedures shall apply:

(a) 25 readout units shall be mounted on a vertical phenolic panel, matrixed in 5 horizontal rows of 5 readouts per row.
(b) Tests shall be conducted at an ambient temperature of $50^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$ C.
(c) At the completion of the test, the five readouts in the fourth row from the bottom shall be tested in accordance with 4.8.4 and 4.8.5.

5. PREPARATION FOR DELIVERY

NOTE: The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Readouts shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Readouts shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Readouts shall be individually packaged in accordance with MIL-P-116, submethod IIc insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Each unit package shall be placed in a supplementary container conforming to PPP-B-566, PPP-B-676 or PPP-B-636, class weather resistant.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Cleaned and dried readouts shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1). Packing may be omitted for unit packages utilizing a container conforming to PPP-B-636, provided the requirements for closure, waterproofing and banding, applicable to the level of packing, are met.

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5.2.1 Level A. The packaged readouts shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged readouts shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. For Army procurement, fiberboard containers shall be class weather-resistant as specified in level A.

5.2.3 Level C. The packaged readouts shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Readouts, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Readouts, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic. For Army procurement, caps shall be weather resistant as specified in level A.

5.2.4.3 Level C. Readouts, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, supplementary and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Ordering data. Readouts (see 3.2) shall be described as follows:

6.1.1 Category I readouts (items covered by specification sheets, see 3.2.1). The procurement document shall specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet and the military part number.
(c) Style and class required (see 1.2).
(d) Whether independent or combination mounting is required (see 3.5.1).

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(e) Voltage and illumination requirements.
(f) Type and quantity of lamps required (see 3.5.8).
(g) Type, style, color, height, and position of characters and/or word groups (see 3.7 and 3.8).
(h) Level of preservation and packaging, packing and applicable marking (see section 5).

6.1.2 Category II readouts (qualified readouts with modifications, see 3.2.2). The procurement document shall specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable specification sheet.
(c) Military part number of qualified readout.
(d) Manufacturer's part number of modified readout (see 3.22 and note 2).
(e) Details of the variations from the specification sheet.
(f) Inspection requirements - (To verify variations from category I readouts) (see 4.6 and note 1).
  (1) Tests to be performed (if any).
  (2) The laboratory at which inspection is to be performed.
  (3) Samples and submission of data, if other than that specified.

Note 1: Available manufacturing test data showing compliance may be substituted as meeting these requirements at option of procuring activity.

Note 2: A copy of the drawing furnished under (e) including the description of the variations from the specification sheet, shall be sent to the preparing activity as listed on the individual specification sheet. Preparation and submission of data shall be in accordance with step II, MIL-STD-749.

6.1.3 Category III readouts (items not covered by specification sheets) (see 3.2.3). For readouts not covered by specification sheets, the procurement document may specify the following:

(a) Title, number, and date of this specification.
(b) Classification (see 1.2.1 to 1.2.2 inclusive).
(c) Part number (see 3.22).
(d) Details of design, construction, physical dimensions, and weight (see 3.5).
(e) Whether independent or combination mounting is required (see 3.5.1).
(f) Terminal design and construction (see 3.5.6).
(g) Terminal marking (see 3.5.7).
(h) Voltage and illumination requirements.
(i) Type and quantity of lamps required (see 3.5.8).
(j) Type, style, color, height, and position of characters and/or word groups (see 3.7 and 3.8).
(k) Other plastic material, if required (see 3.4.1).
(l) Other finish, if required (see 3.4.7).
(m) Inspection requirements, if other than that specified in 3.2 and 4.6 and
  (1) The laboratory at which the inspection is to be performed (see 4.6).
  (2) Samples and submission of data, if other than that specified (see 4.6).

Note: Available manufacturing test data showing compliance may be substituted as meeting these requirements, at option of the procuring activity.
(n) Character display (see 4.8.3).
(o) Display luminance (see 4.8.6).
(p) Field of view (see 4.8.7).
(q) Terminal strength (see 4.8.9).
(r) Shock method of other than method I (see 4.8.12).
(s) Vibration method and test condition letter if other than method I (see 4.8.15).
(t) Level of preservation and packaging, packing and applicable marking (see section 5).

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6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Electronic Systems Command, Department of the Navy, Washington, D.C. 20360; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45401.

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.3 Grouping of metals (see 3.4.4). The grouping in table VI is intended to serve as a guide in selecting materials to be used in electronic equipment, and should not be construed to waive requirements herein or in the applicable specification sheet as to corrosion resistance of components and assemblies.

6.3.1 Application. Unless specifically approved by the Government, all metals not listed in table VI should be considered dissimilar with respect to one another and with respect to any of the materials listed. Except for zinc, cadmium, and tin as listed in groups II and III, and for stainless steel as listed in groups II, III, and IV, contact between a member of one group and a member of another group should be considered dissimilar. Such contact should not be made unless necessary, in which case it should be demonstrated that the contact is not detrimental, or an interposing material should be used which will be compatible to each metal. When reference is made to a metal in a particular group, the reference applies to the metal on the surface of the part; i.e., zinc means zinc coating as well as zinc electroplate, zinc hotdip, or zinc metal-spray. Different metals in contact, even though similar, should be employed in assemblies in such a manner that the smaller part is cathodic (or protected) and the larger part is anodic (or corroded) if any corrosion takes place. Care should be exercised in using different aluminum alloys against each other or against differing material.

TABLE VI. Grouping of metals.

Group I	Group II	Group III	Group IV
Magnesium alloys (most anodic)	Aluminum Aluminum alloys Zinc Cadmium Tin Stainless steel	Zinc Cadmium Steel Lead Tin Stainless steel	Copper and its alloys Nickel and its alloys Chromium Stainless steel Gold Silver (most cathodic)

6.4 Definitions.

Readout projection unit - Projects a message, which has been photographed on film and mounted in front of a lamp, through a projection lens, and displaying its magnified image upon a viewing screen.

Readout, backlighted unit - Backlights with a lamp directly behind a message position, and illuminating, without magnification, the message affixed to the viewing screen.

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6.5 Application notes.

6.5.1 Selection of colors (see 3.8.2). In general, the selection of colors should conform to the following:

(a) Red - Red lights should display information of an unsatisfactory condition.
(b) Green - Green lights should display information of satisfactory condition.
(c) Amber - Amber should display information of a caution or marginal condition.
(d) White - White should designate system condition or status, should not be used to imply a right or wrong condition.

In general, whatever color code is selected, it should be consistent throughout the system. Blue color is not recommended.

6.5.2 Lettering (see 3.8.1). Lettering should conform to the following:

(a) If abbreviations are used, MIL-STD-12 "Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents", should be used as a guide.
(b) All lettering should be capitals. Lettering should not be smaller than 1/8-inch in height.
(c) Use Arabic numerals instead of Roman.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI
Navy - AS, OS
Air Force - 17, 82
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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MIL-R-39027A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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